Task Space Compliant Control and Six-Dimensional Force Regulation Toward Automated Robotic Ultrasound Imaging

Junchen Wang, Chunheng Lu, Yifei Lv, Siqin Yang, Mingbo Zhang, and Yu Shen

***Abstract*— *Objective:* We propose a general control framework for task space compliant motion and six-dimensional (6-D) force regulation towards automated robotic ultrasound (US) imaging. The framework endows a position-controlled robotic manipulator with the capability of accurate compliant motion in free space and accurate force control in motion-constrained environment. *Methods:* An intuitive six degree-of-freedom (6-DoF) admittance control model expressed in an arbitrary Cartesian body frame is mathematically derived with closed-form task space error mapping. Its practical implementation on widely-used collaborative manipulators is proposed to achieve full task space compliant behaviors and accurate 6-D force control. A hybrid control law is presented to achieve good motion accuracy in free space and improved coupled stability in motion-constrained environment. The coupled model of physical human-robot interaction is established and the reason for the improved coupled stability is analyzed through simulation. *Results:* Evaluation experiments on the proposed control framework were performed to show the effectiveness. The mean error of compliant trajectory following was less than 0.30 mm in free space. The mean relative force and moment control accuracy in three orthogonal directions was better than 0.5% and 0.8%, respectively. The improved coupled stability under the same model parameters was also confirmed by human-robot interaction experiments. Finally, an automated robotic US imaging experiment on a human volunteer in a real clinical scenario was carried out to show the potential application of our proposed framework. *Conclusion:* Experimental results have shown the advantages of the control framework, including satisfied force control accuracy, high accuracy of compliant motion, improved coupled stability, and system effectiveness on a human volunteer.**

***Note to Practitioners*—This paper was motivated by the increasing needs of automated ultrasound (US) scanning for both diagnostic and interventional purpose. Clinical sonographers suffer from repeated workload when performing diagnostic US imaging, which could benefit from automated robotic scanning. Robotic US imaging involves physical interaction between the robot end-effector (i.e., US transducer) and the human body. The dynamics of the interaction is regulated by the control law to guarantee the contact of the US transducer and the safety of the procedure. Most existing works have focused on regulating in-plane contact force in terms of the position without considering the compliance in other dimensions. However, it is not a trivial work to extend the positional compliance to six degree-of-freedom (6-DoF) compliance. As the prevalence of low-cost collaborative robotic arms in medical scenarios, how to perform 6-DoF compliant trajectory following and accurate six-dimensional (6-D) force control on these robotic arms becomes increasingly important. This paper gives a complete general solution to achieve 6-DoF compliant control and 6-D force regulation with accurate kinematics on a position-controlled robotic arm. A hybrid control law is proposed to switch the government of "instantaneous model" and "theoretical model" to achieve compliant motion accuracy in free space and improved coupled stability in motion-constrained environment. No expensive torque sensors and torque control interface are required. And no prior geometric knowledge about the scanning object is needed. We have demonstrated the application for robotic US imaging in a real clinical scenario.**

***Index Terms*— Robotic ultrasound, admittance control, force control, coupled stability.**

I. Introduction

ROBOTIC ultrasound (US) imaging has been proposed and actively researched in the last decade [1], [2], [3], [4], in which a US transducer (probe) is attached to a robotic arm to increase scan accuracy, motion repeatability and image interpretation. Powered by the robotic control technology, the robotic US imaging has been making remote diagnosis and autonomous scanning possible [5], [6], [7], [8], [9], which could significantly reduce physical burdens of sonographers in clinical scan procedures and avoid radiation/infectious risks in intraoperative US imaging procedures [10], [11]. Compared with other image modalities such as CT or MRI, US imaging has advantages of low cost, real-time imaging and non-radiation, which makes it prevalent in clinical diagnosis. However, it suffers from difficulties in image interpretation and the image quality highly relies on the interaction (motion and force) between the probe and the human body. Physical contact with an appropriate pressure force between the probe and the human body should be maintained during the image acquisition. In addition, the position and orientation of the probe is frequently adjusted to image different targets of internal structures. In the case of a human sonographer, safe

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and effective interaction is guaranteed by human intelligence and well-trained professional skills. Robotic US imaging requires compliant robot behaviors to guarantee safe interaction between the robot end-effector (i.e., US probe) and the human body. Pure position controlled robots are not appropriate for robotic US imaging since the task geometry constraint causes damages to either the robot or the human body.

Compliance can be divided into two categories: passive and active compliance. Passive compliance is achieved by variable stiffness elements or adaptable soft material structures. As the development of soft robotics, adaptable grippers with soft materials have drawn great attention in the robotics community, which can be used to grasp irregular sized objects with compliant behaviors [12], [13], [14]. For robotic US imaging, Bao et al. [15] developed a 1-DoF constant-force end-effector to adjust the vertical contact force of the probe. The end-effector adopts a motor-spring-based structure to generate constant operating force in vertical direction. Due to the limited operating space (vertical displacement within several millimeters), the general motion of the probe is provided by the robot arm. In addition, the end-effector employs step motors without position encoders which brings difficulties in retrieving accurate kinematic data. Lindenroth et al. [16] proposed a parallel soft robotic end-effector for extracorporeal US imaging. Soft fluidic actuators were employed to establish compliant interaction between the US probe and the patient. The position control accuracy of the probe was achieved using an external tracking device which adversely affects it clinical application. Besides, small workspace and additional large driving units are also the limits of the work. Although the passive compliance has good adaptation with simple control strategy, it suffers from inaccurate kinematics and large peripheral driving units. One of the promising advantages of robotic US imaging is to reconstruct a 3D volume from 2D US images with pose information to enhance the spatial perception towards the scanning area [17]. This requires accurate kinematic chain from the robot's base frame to the US probe.

Active compliance on a rigid robot has advantages of no need to significantly modify/add the hardware and no loss in kinematic accuracy [18]. Common active compliance control includes force/position control, impedance control and admittance control [19], [20]. Actually, impedance control and admittance control aim at implementing the same control goal but with different approaches. In impedance control, the robot senses the error between actual and desired motion and renders resulting force to the environment. While in admittance control, the robot senses the force between the robot and the environment and renders resulting motion. Impedance control requires that the robot can be controlled with joint torque inputs. It inherently depends on an accurate dynamic model of the robot and is very sensitive to the model uncertainties [21], [22]. Since the admittance control can be applied on a force sensor equipped robot with position control interface, it is possible to combine accurate motion control and compliant interaction to achieve safe robotic US imaging on a low-cost collaborative robot manipulator [23].

With regard to robotic US imaging, most existing works have focused on regulating the contact force between the probe and the human body in particular dimensions without considering other dimensions (task space up to six dimensions). Moreover, the force regulation process is often coupled with specific scene features thus loses generality. Huang et al. [24], [25], [26] attached two 1-D force sensors to the probe's emitting plane and used a simple strategy to adjust the probe's in-plane motion. This method relies on accurate 3D scanning data of the contact surface which thereby loses generality. Li et al. [27] proposed a robotic spinal sonography system using the KUKA iiwa robot with Cartesian impedance control, which requires the robot to have an accurate torque control interface. Victorova et al. [28] converted measured force error along the normal direction to the velocity and controlled the velocities along other two orthogonal directions. This method simply assumes that the scan workspace is a cube and the probe is strictly perpendicular to the body surface during the scan. Ma et al. [29] presented a 1-D force-distance mapping method to place the probe against the human body with desired contact force. The method relies on external sensing to obtain geometries of the target to align the direction of force-distance mapping with the normal of the contact surface. Huang et al. [30] proposed a US image feature-based visual servo controller for carotid artery robotic US scanning. The controller maps the 6-D force to 6-D velocity twist using a linear transform to regulate the contact force. Due to the mapping is performed in-plane, the out-plane force regulation is lost. Jiang et al. [31] and Welleweerd et al. [32] employed a torque-controlled KUKA iiwa manipulator to ensure safe interaction during the robotic scanning. However, the force control accuracy was not presented.

To conclude, almost all existing works (to the best of our knowledge) regulates the contact force of robotic US end-effectors along a predefined direction. This direction usually refers to the normal direction of contact surface. The resulting compliant behavior only spans the in-plane space of the US probe. We think the in-plane compliance is not safe enough for robotic US imaging in complicated clinical scenarios. For example, if the patient suddenly moves his body to block the motion direction of the robot which is outside the compliant space, large contact force occurs. The preferred behavior is that the robot's motion should be compliant with the unexpected obstruction from all possible directions. Moreover, it lacks a general control framework for accurate 6-D force regulation which does not rely on any specific scene or prior geometry knowledge in the sate-of-the-art robotic US imaging systems. The framework is supposed to be implemented on low-cost robot manipulators with position control interface. Any specific feature-based servo algorithm can be directly overlaid.

In this paper, we propose such a general control framework for 6-DoF task space compliant motion and accurate 6-D force regulation towards automated robotic US imaging. The main contribution of our work is summarized as follows.

- A general geometrically meaningful 6-DoF admittance control model with closed-form task space error mapping.

- Practical implementation with feedforward motion estimation and feedback re-initialization that enables both accurate compliant trajectory following and interaction.
- Accurate 6-D force control model for physical human-robot interaction with improved coupled stability.

II. Methods

A. Notation and Problem Statement

We denote by $[\boldsymbol{\omega}]$ the skew-symmetric matrix representation of a vector $\boldsymbol{\omega} \in \mathbb{R}^3$ and define the 4×4 matrix representation of a vector $\boldsymbol{\mathcal{V}} = (\boldsymbol{\omega}; \boldsymbol{v}) \in \mathbb{R}^6$ as

$$[\boldsymbol{\mathcal{V}}] = \begin{pmatrix} [\boldsymbol{\omega}] & \boldsymbol{v} \\ \boldsymbol{0} & 0 \end{pmatrix} \tag{1}$$

If we look at $\boldsymbol{\mathcal{V}}$ as a twist and re-write it as $\boldsymbol{\mathcal{V}} = \theta \boldsymbol{\mathcal{S}}$ where $\boldsymbol{\mathcal{S}} = \left(\hat{\boldsymbol{\omega}}; \hat{\boldsymbol{v}}\right)$ is the screw axis so that $\|\hat{\boldsymbol{\omega}}\| = 1$ or $\hat{\boldsymbol{\omega}} = \boldsymbol{0} \wedge \|\hat{\boldsymbol{v}}\| = 1$, the matrix exponential of $[\boldsymbol{\mathcal{V}}]$ is a 4×4 homogeneous transformation matrix given by

$$e^{[\boldsymbol{\mathcal{V}}]} = e^{[\theta \boldsymbol{\mathcal{S}}]} = \begin{pmatrix} e^{[\theta\hat{\boldsymbol{\omega}}]} & \left(\theta \boldsymbol{I}_3 + (1-\cos\theta)\left[\hat{\boldsymbol{\omega}}\right] + (\theta - \sin\theta)\left[\hat{\boldsymbol{\omega}}\right]^2\right)\hat{\boldsymbol{v}} \\ \boldsymbol{0} & 1 \end{pmatrix} \tag{2}$$

where $\boldsymbol{I}_3 \in \mathbb{R}^{3\times 3}$ is an identity matrix and $e^{[\theta\hat{\boldsymbol{\omega}}]}$ is given by the Rodrigues's rotation formula. We further define the log operator on a rotation matrix $\boldsymbol{R} \in SO(3)$ that returns its rotation vector $\boldsymbol{r} \in \mathbb{R}^3$ so that the matrix exponential $e^{[\boldsymbol{r}]} = \boldsymbol{R}$ holds. Given a 4×4 homogeneous transformation matrix $\boldsymbol{T} \in SE(3)$, its adjoint representation $[\mathrm{Ad}_{\boldsymbol{T}}] \in \mathbb{R}^{6\times 6}$ is defined as

$$[\mathrm{Ad}_{\boldsymbol{T}}] = \begin{pmatrix} \boldsymbol{R} & \boldsymbol{0} \\ [\boldsymbol{p}]\boldsymbol{R} & \boldsymbol{R} \end{pmatrix} \tag{3}$$

where $\boldsymbol{R}$ and $\boldsymbol{p}$ are the rotation and translation part of $\boldsymbol{T}$. For simplicity, we use $\boldsymbol{T} = (\boldsymbol{R}, \boldsymbol{p})$ to represent a homogeneous transformation matrix with $\boldsymbol{R}$ and $\boldsymbol{p}$ being the rotation and translation part, respectively. $[\mathrm{Ad}_{\boldsymbol{T}}]$ is used to relate the velocity twists of a rigid object expressed in different frames, namely, $\boldsymbol{\mathcal{V}}_1 = [\mathrm{Ad}_{\boldsymbol{T}_{12}}]\boldsymbol{\mathcal{V}}_2$ where $\boldsymbol{\mathcal{V}}_1$ and $\boldsymbol{\mathcal{V}}_2$ are the twists of the object expressed in frame $\{1\}$ and $\{2\}$, respectively, and $\boldsymbol{T}_{12}$ is the transformation matrix of frame $\{2\}$ with respect to frame $\{1\}$.

The general purpose of both impedance and admittance control is to achieve the following dynamics of the robot's end-effector:

$$\boldsymbol{M}\ddot{\boldsymbol{x}}_e + \boldsymbol{B}\dot{\boldsymbol{x}}_e + \boldsymbol{K}\boldsymbol{x}_e = \boldsymbol{\mathcal{F}} \tag{4}$$

where $\boldsymbol{M}, \boldsymbol{B}, \boldsymbol{K} \in \mathbb{R}^{6\times 6}$ are the mass, damping and stiffness matrices, respectively; $\boldsymbol{x}_e \in \mathbb{R}^6$ is the 6-DoF pose error of the robot in task space; and $\boldsymbol{\mathcal{F}} \in \mathbb{R}^6$ is the external wrench experienced or exerted by the robot depending on the representation of $\boldsymbol{x}_e$. Eq. (4) is frequently seen in the literature, but it is just an abstraction model whose concrete form depends on both the representation of $\boldsymbol{x}_e$ and the expression frame [33]. For clarity, we define the notations summarized in Table I and illustrated in Fig. 1 for a geometrically meaningful model derivation. Unless indicated explicitly, motion variables are represented in the global frame of the robot. Note that sometimes we will drop the dependence on time t for simplicity.

TABLE I
NOTATIONS AND DESCRIPTIONS

n	degrees of freedom of the robot manipulator
$\{\mathrm{w}\}$	global frame of the robot
$\{\mathrm{b}\}$	body frame rigidly attached to the robot's end-effector
$\{\mathrm{s}\}$	spatial frame whose rotation is an identity matrix and origin coincides with that of $\{\mathrm{b}\}$
$\boldsymbol{q}(t) \in \mathbb{R}^n$	joint angles of the robot
$\boldsymbol{J}_b(t) \in \mathbb{R}^{6\times n}$	Jacobian matrix expressed in $\{\mathrm{b}\}$
$\boldsymbol{T}_d(t) \in SE(3)$	desired trajectory of $\{\mathrm{b}\}$
$\boldsymbol{R}_d(t) \in SO(3)$	rotation part of $\boldsymbol{T}_d(t)$
$\boldsymbol{p}_d(t) \in \mathbb{R}^3$	translation part of $\boldsymbol{T}_d(t)$
$\boldsymbol{\omega}_d(t) \in \mathbb{R}^3$	desired angular velocity of $\{\mathrm{b}\}$
$\boldsymbol{v}_d(t) = \dot{\boldsymbol{p}}_d(t)$	desired linear velocity of $\{\mathrm{b}\}$
$\boldsymbol{\mathcal{V}}_d(t) = (\boldsymbol{\omega}_d(t); \boldsymbol{v}_d(t))$	desired spatial twist of $\{\mathrm{b}\}$
$\boldsymbol{T}(t) \in SE(3)$	actual trajectory of $\{\mathrm{b}\}$
$\boldsymbol{R}(t) \in SO(3)$	rotation part of $\boldsymbol{T}(t)$
$\boldsymbol{p}(t) \in \mathbb{R}^3$	translation part of $\boldsymbol{T}(t)$
$\boldsymbol{\omega}(t) \in \mathbb{R}^3$	angular velocity of $\{\mathrm{b}\}$
$\boldsymbol{v}(t) = \dot{\boldsymbol{p}}(t)$	linear velocity of $\{\mathrm{b}\}$
$\boldsymbol{\omega}_b(t) = \boldsymbol{R}^{\mathrm{T}}(t)\boldsymbol{\omega}(t)$	angular velocity of $\{\mathrm{b}\}$ expressed in $\{\mathrm{b}\}$
$\boldsymbol{v}_b(t) = \boldsymbol{R}^{\mathrm{T}}(t)\dot{\boldsymbol{p}}(t)$	linear velocity of $\{\mathrm{b}\}$ expressed in $\{\mathrm{b}\}$
$\boldsymbol{\mathcal{V}}(t) = (\boldsymbol{\omega}(t); \boldsymbol{v}(t))$	spatial twist of $\{\mathrm{b}\}$
$\boldsymbol{\mathcal{V}}_b(t) = (\boldsymbol{\omega}_b(t); \boldsymbol{v}_b(t))$	body twist of $\{\mathrm{b}\}$
$\boldsymbol{\mathcal{F}} = \left(\boldsymbol{\mathcal{F}}^m; \boldsymbol{\mathcal{F}}^f\right) \in \mathbb{R}^6$	wrench with $\boldsymbol{\mathcal{F}}^m$ representing moments and $\boldsymbol{\mathcal{F}}^f$ representing forces

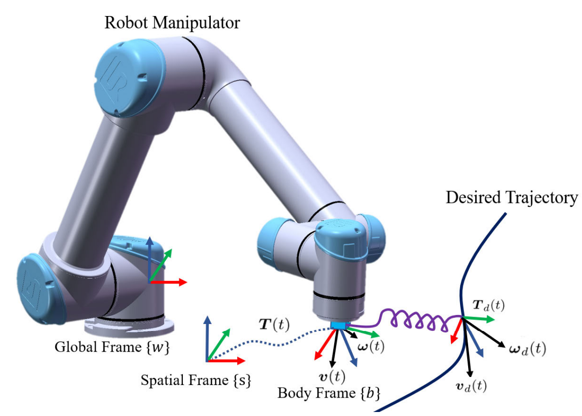

Fig. 1. Frame and velocity illustration of a compliant robot.

To represent Eq. (4) in the body frame $\{\mathrm{b}\}$, $\boldsymbol{x}_e$ is chosen to be $\boldsymbol{x}_e = \left(\boldsymbol{r}_e; \boldsymbol{p}_e\right)$ where

$$\boldsymbol{r}_e = \log \boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}_d \tag{5}$$

$$\boldsymbol{p}_e = \boldsymbol{R}^{\mathrm{T}}\left(\boldsymbol{p}_d - \boldsymbol{p}\right) \tag{6}$$

With the representation of $\boldsymbol{M} = \mathrm{diag}(\boldsymbol{M}_r, \boldsymbol{M}_p)$, $\boldsymbol{B} = \mathrm{diag}(\boldsymbol{B}_r, \boldsymbol{B}_p)$, $\boldsymbol{K} = \mathrm{diag}(\boldsymbol{K}_r, \boldsymbol{K}_p)$, Eq. (4) can be decoupled into the rotational and translational dynamics as

$$\boldsymbol{M}_r\ddot{\boldsymbol{r}}_e + \boldsymbol{B}_r\dot{\boldsymbol{r}}_e + \boldsymbol{K}_r\boldsymbol{r}_e = \boldsymbol{\mathcal{F}}_b^m \tag{7}$$

$$\boldsymbol{M}_p\ddot{\boldsymbol{p}}_e + \boldsymbol{B}_p\dot{\boldsymbol{p}}_e + \boldsymbol{K}_p\boldsymbol{p}_e = \boldsymbol{\mathcal{F}}_b^f \tag{8}$$

where $\left(\boldsymbol{\mathcal{F}}_b^m; \boldsymbol{\mathcal{F}}_b^f\right)$ is the wrench **exerted** on the environment by the robot expressed in $\{\mathrm{b}\}$. Eq. (7) has a clear geometrical meaning. If a moment $\boldsymbol{K}_r^{-1}\mathcal{F}_b^m$ is **applied** to the body frame which is on its desired pose, after equilibrium it will deviate from the **desired** pose by performing a corresponding rotation represented by the rotation vector $-\boldsymbol{r}_e = \boldsymbol{K}_r^{-1}\mathcal{F}_b^m$ with respect to itself. Since the resulting rotation is consistent with the applied moment, the compliant motion is intuitive or geometrically meaningful. The geometrical meaning of Eq. (8) is straightforward. Compared with expressing Eq. (4) in the spatial frame {s}, the impedance characteristic does not rely on the orientation of {b} and is completely determined by the parameters $\boldsymbol{M}, \boldsymbol{B}, \boldsymbol{K}$.

B. Ideal Admittance Controller

Assume an ideal position-controlled robot which is controlled continuously without trajectory tracking error. A wrist force/torque sensor is used to measure the external wrench $\mathcal{F}_b(t)$. The sensed wrench is subject to a second order system (4) to be converted to the pose errors $\boldsymbol{r}_e(t)$ and $\boldsymbol{p}_e(t)$, which are used to modify the desired trajectory $\boldsymbol{T}_d(t)$. The resulting trajectory $\boldsymbol{T}(t)$ is commanded to a position-controlled robot to achieve the desired dynamics (7) and (8). Since the admittance controller does not use any feedback data from the robot and assumes the perfect position control, it is called the ideal admittance controller. The ideal controller requires that the commanded $\boldsymbol{T}(t)$ can be accurately achieved without delay. However, in a practical system: 1. Control cycle is not infinitesimal; 2. Physical delay occurs; 3. Trajectory tracking is subject to error. These practical limits degrade the performance of the ideal controller and lead to coupled instability as we shall show. Its advantage is that the commanded trajectory $\boldsymbol{T}(t)$ quickly converges to $\boldsymbol{T}_d(t)$ when contact force disappears, provided appropriate parameters of $\boldsymbol{M}, \boldsymbol{B}, \boldsymbol{K}$. This means that the compliant robot will return to the desired trajectory when no external wrench is applied. Therefore, the ideal controller has accurate trajectory following performance in free space.

C. Practical Admittance Controller

To avoid accumulated error in numerical integration, a practical admittance controller with feedforward motion estimation and feedback re-initialization is proposed, which takes the robot's joint angles and speeds as feedback data. The core of the practical controller is to derive the mapping from the robot's state $(\boldsymbol{q}, \dot{\boldsymbol{q}})$ in joint space to the error state $(\boldsymbol{x}_e, \dot{\boldsymbol{x}}_e)$ in task space, given $\boldsymbol{T}_d$ and $\mathcal{V}_d$. $\boldsymbol{x}_e$ can be directed calculated using Eq. (5) and (6) with $(\boldsymbol{R}, \boldsymbol{p}) = FK(\boldsymbol{q})$, where $FK(\boldsymbol{q})$ represents the forward kinematics of the robot. The body twist $\mathcal{V}_b = (\boldsymbol{w}_b; \boldsymbol{v}_b)$ is calculated using $\mathcal{V}_b = \boldsymbol{J}_b\dot{\boldsymbol{q}}$, which is related to the spatial twist $\mathcal{V} = (\boldsymbol{w}; \boldsymbol{v})$ by

$$\mathcal{V} = \begin{pmatrix} \boldsymbol{R} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{R} \end{pmatrix} \mathcal{V}_b \tag{9}$$

Let $\boldsymbol{r} = \log \boldsymbol{R}$, the angular velocity $\boldsymbol{\omega}_b$ (note that $[\boldsymbol{w}_b] = \boldsymbol{R}^{-1}\dot{\boldsymbol{R}}$) is related to $\dot{\boldsymbol{r}}$ by

$$\boldsymbol{\omega}_b = \boldsymbol{A}(\boldsymbol{r})\dot{\boldsymbol{r}} \tag{10}$$

where

$$\boldsymbol{A}(\boldsymbol{r}) = \boldsymbol{I}_3 - \frac{1-\cos\|\boldsymbol{r}\|}{\|\boldsymbol{r}\|^2}[\boldsymbol{r}] + \frac{\|\boldsymbol{r}\| - \sin\|\boldsymbol{r}\|}{\|\boldsymbol{r}\|^3}[\boldsymbol{r}]^2 \tag{11}$$

The proof of Eq. (10) and (11) is given in **Appendix**. Taking the skew matrix of both sides of Eq. (10) and making substitutions of $\boldsymbol{r} \leftarrow \boldsymbol{r}_e$, $\boldsymbol{R} \leftarrow \boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}_d$, we have

$$\begin{aligned} \left[\boldsymbol{A}(\boldsymbol{r}_e)\dot{\boldsymbol{r}}_e\right] &= \left(\boldsymbol{R}^{-1}\boldsymbol{R}_d\right)^{-1}\left(\boldsymbol{R}^{-1}\dot{\boldsymbol{R}}_d - \boldsymbol{R}^{-1}\dot{\boldsymbol{R}}\boldsymbol{R}^{-1}\boldsymbol{R}_d\right) \\ &= \left[\boldsymbol{R}_d^{-1}\boldsymbol{\omega}_d - \boldsymbol{R}_d^{-1}\boldsymbol{\omega}\right] \end{aligned} \tag{12}$$

based on the facts of $\boldsymbol{R}^{\mathrm{T}} = \boldsymbol{R}^{-1}$, $\dot{\left(\boldsymbol{R}^{-1}\right)} = -\boldsymbol{R}^{-1}\dot{\boldsymbol{R}}\boldsymbol{R}^{-1}$, $\dot{\boldsymbol{R}}_d\boldsymbol{R}_d^{-1} = [\boldsymbol{w}_d]$, $\dot{\boldsymbol{R}}\boldsymbol{R}^{-1} = [\boldsymbol{w}]$, $\boldsymbol{R}^{-1}\dot{\boldsymbol{R}} = [\boldsymbol{w}_b]$. Finally, we have

$$\dot{\boldsymbol{r}}_e = \boldsymbol{A}^{-1}(\boldsymbol{r}_e)\boldsymbol{R}_d^{-1}(\boldsymbol{\omega}_d - \boldsymbol{\omega}) \tag{13}$$

Similarly, differentiating both sides of Eq. (6) with respect to time t, we have

$$\begin{aligned} \dot{\boldsymbol{p}}_e &= \boldsymbol{R}^{-1}\left(\dot{\boldsymbol{p}}_d - \dot{\boldsymbol{p}}\right) - \boldsymbol{R}^{-1}\dot{\boldsymbol{R}}\boldsymbol{R}^{-1}\left(\boldsymbol{p}_d - \boldsymbol{p}\right) \\ &= \boldsymbol{R}^{-1}(\boldsymbol{v}_d - \boldsymbol{v}) - [\boldsymbol{\omega}_b]\boldsymbol{p}_e \end{aligned} \tag{14}$$

By far we have calculated the current error state $\boldsymbol{x}_e = (\boldsymbol{r}_e; \boldsymbol{p}_e)$ and $\dot{\boldsymbol{x}}_e = (\dot{\boldsymbol{r}}_e; \dot{\boldsymbol{p}}_e)$ given the joint state and desired motion, which we call the task space error mapping. The dynamics (4) tells us how to update the error state at next step:

$$\ddot{\boldsymbol{x}}_e^k = \boldsymbol{M}^{-1}\left(\mathcal{F}_b^k - \boldsymbol{K}\boldsymbol{x}_e^k - \boldsymbol{B}\dot{\boldsymbol{x}}_e^k\right) \tag{15}$$

$$\dot{\boldsymbol{x}}_e^{k+1} = \dot{\boldsymbol{x}}_e^k + \ddot{\boldsymbol{x}}_e^k\delta t \tag{16}$$

$$\boldsymbol{x}_e^{k+1} = \boldsymbol{x}_e^k + \dot{\boldsymbol{x}}_e^{k+1}\delta t \tag{17}$$

where δt is the control cycle. The desired pose $\boldsymbol{T}_d^{k+1}$ at next step is estimated by

$$\boldsymbol{T}_d^{k+1} = \boldsymbol{T}_d^k e^{\left[[\mathrm{Ad}_{\boldsymbol{T}_{bs}^k}]\mathcal{V}_d^k\delta t\right]} \tag{18}$$

where the exponential is the 4×4 matrix representation of the screw motion $[\mathrm{Ad}_{\boldsymbol{T}_{bs}^k}]\mathcal{V}_d^k\delta t$, and $\boldsymbol{T}_{bs}^k$ is a homogeneous transformation given by

$$\boldsymbol{T}_{bs}^k = \begin{pmatrix} \left(\boldsymbol{R}_d^k\right)^{-1} & \boldsymbol{0} \\ \boldsymbol{0} & 1 \end{pmatrix} \tag{19}$$

With the feedforward motion estimation $\boldsymbol{T}_d^{k+1}$, the commanded robot's task pose is given by

$$\boldsymbol{R}^{k+1} = \boldsymbol{R}_d^{k+1}e^{[-\boldsymbol{r}_e^{k+1}]} \tag{20}$$

$$\boldsymbol{p}^{k+1} = \boldsymbol{p}_d^{k+1} - \boldsymbol{R}^{k+1}\boldsymbol{p}_e^{k+1} \tag{21}$$

The commanded joint position $\boldsymbol{q}^{k+1}$ is resolved from $\boldsymbol{T}^{k+1} = (\boldsymbol{R}^{k+1}, \boldsymbol{p}^{k+1})$ by inverse kinematics. The implementation of the practical admittance controller is illustrated in Fig. 2. As we shall show the practical controller has improved coupled stability, its disadvantage is inaccurate trajectory following in free space. This is because in practice the commanded motion cannot be accurately achieved within a control cycle, which causes lasting non-zero errors of both $\boldsymbol{x}_e$ and $\dot{\boldsymbol{x}}_e$ even if no external wrench presents.

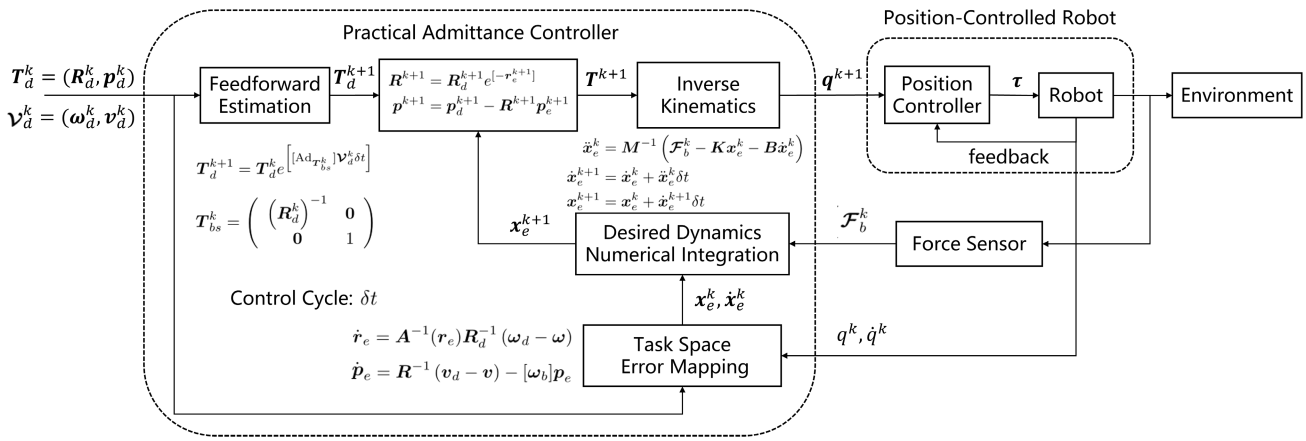

Fig. 2. Practical admittance controller.

D. Coupled Stability Analysis

Coupled stability refers to the stability when the robot is in a motion-constrained environment such as physically interacting with the human body or held by the human. Two separately stable systems can exhibit coupled instability. For controlled devices interacting with a human user, the stability behavior is therefore highly dependent on the user's impedance characteristics. A sufficient condition to ensure coupled stability is that the controlled robot's dynamics are energetically passive. The theoretical coupled stability analysis of admittance control has been well studied and discussed in the literature such as in [23]. For example, increasing the damping coefficient can improve the coupled stability at the cost of rendering a heavy interaction feeling. However, we do not intend to discuss the parameter choosing rules for the trade-off between the interaction and stability. We focus on the coupled stability considering the "implementation limits".

The difference between the ideal and practical controller lies in whether re-initializing the error state $(\boldsymbol{x}_e, \dot{\boldsymbol{x}}_e)$ according to the measured joint angles/speeds when numerically solving the ordinary differential equation (4). The underlying theoretical models are the same given the same model parameters but with different practical implementation. The ideal controller starts with an initial error state $(\boldsymbol{x}_e^0, \dot{\boldsymbol{x}}_e^0)$ and the following evolution is completely governed by the equation (4) in spite of the actual state of the robot. It assumes perfect position control of the robot. Actually, the commanded motion from the inner position controller according to the theoretical control law cannot be achieved within a control cycle. This leads to large accumulated error when preforming the numeric integration to solve the differential equation of the dynamics. This actually changes the designed dynamics and leads to a time-varying impedance characteristic that does not meet the passivity. However, in the practical controller, the time-varying impedance effect is alleviated and rectified by measuring the state of the robot at each control cycle.

Under the same model parameters, we shall prove the improved coupled stability of the practical controller against the ideal controller by considering the actual joint position tracking performance. Fig. 3 shows the position tracking performance of the shoulder joint of a UR5e robot at a speed of 0.05 rad/s. This speed is even below the speed of normal interaction. In Fig. 3(a), the tracking performance index (TPI) is defined as $TPI = (q^{k+1} - q^k)/(q_c^k - q^k)$, where q^k and q^{k+1} represent the actual joint position at k and $k+1$-th control cycle, respectively. q_c^k represents the commanded joint position at k-th control cycle. If the tracking is perfect, the commanded q_c^k at k-th cycle will be achieved at $k+1$-th cycle, i.e., $q^{k+1} = q_c^k$, which means the tracking performance index approaches 1.0. Fig. 3(a) indicates the mean tracking performance index was less than 0.1. The red and blue line in Fig. 3 (b) represent the commanded and actual position of the joint with respect to the control cycle (2 ms). As we can see, the tracking latency is approximately 20 ms, which is about 10 times of the control cycle. In another word, the commanded motion (joint displacement) can be only achieved by one tenth within a control cycle.

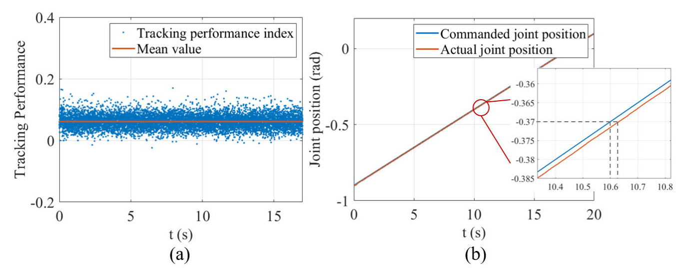

Fig. 3. Tracking performance of the shoulder joint of a UR5e robot. (a) Tracking performance index. (b) The difference between commanded joint positions and actual joint positions.

Based on this fact, we establish a human-robot interaction simulation model as shown in Fig. 4. The robot model is UR5e. The joint position from the admittance controller q_c^k is further subject to a tracking performance index of $TPI = 0.1$ to simulate the imperfect tracking. Since the human and the robot are coupled together (e.g., the robot's end-effector is firmly held by the human), their Cartesian positions are the same. The human impedance characteristics are given by $\boldsymbol{K}_h = \mathrm{diag}(500, 500, 500, 6000, 6000, 6000)$ and $\boldsymbol{B}_h = \mathrm{diag}(3, 3, 3, 10, 10, 10)$ [34]. The admittance controller's parameters are given as $\boldsymbol{M}_r = \mathrm{diag}(0.1, 0.1, 0.1)$,

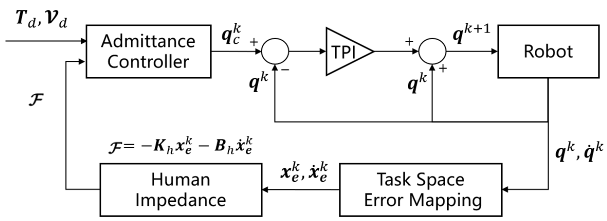

Fig. 4. Coupled stability simulation flow chart.

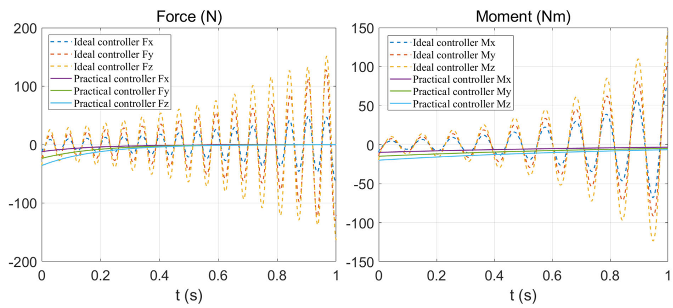

Fig. 5. Simulation Results of interaction force and moment under ideal and practical admittance controllers.

$\boldsymbol{B}_r = \mathrm{diag}(2, 2, 2)$, $\boldsymbol{K}_r = \mathrm{diag}(10, 10, 10)$, $\boldsymbol{M}_p = \mathrm{diag}(0.25, 0.25, 0.25)$, $\boldsymbol{B}_p = \mathrm{diag}(20, 20, 20)$, $\boldsymbol{K}_p = \mathrm{diag}(300, 300, 300)$. The desired motion is a setpoint control with the initial pose error of $\boldsymbol{r}_e^0 = (0.02; 0.03; 0.04)$, $\boldsymbol{p}_e^0 = (2e{-}3; 4e{-}3; 6e{-}3)$. This is mimicking that the human holds the robot's end with an offset from its equilibrium position. All the units are the IS units.

We simulate the model in Fig. 4 with the ideal and practical controller, respectively. The simulation step for numerical integration is 2 ms and the total simulation time is 1 s. The interaction force and moments are recorded to show the coupled stability. The simulation results are shown in Fig. 5. As we can see, the ideal controller presents unstable behavior with the interaction force diverges. The practical controller, on the other hand, has good coupled stability. Please refer to Supplementary Video S1 for more details.

E. Hybrid Admittance Controller

The practical controller is more like rendering an instantaneous impedance model at every control cycle. Its advantage is improved coupled stability due to the feedback, but the disadvantage is inaccurate desired motion tracking even no external wrench is present. However, if no external wrench is present in the ideal controller, $\boldsymbol{x}_e$ exponentially decays to zero provided that appropriate parameters for critical damping are set. Therefore, the ideal and practical controllers have complementary advantages which inspires us to combine them to create a hybrid controller. The idea of the hybrid controller is simply adding a switch to enable re-initialization of the error state from the actual data of the robot. If the norm of the current external wrench $\|\mathcal{F}_c^k\|$ is below a threshold, the switch is off to disable re-initialization (keeping last values). Otherwise, the re-initialization is performed at current control cycle.

F. 6-D Force Control Model

The admittance model (7) and (8) can be easily modified into a 6-D force control model which aims to provide constant force/moment along specified directions in frame $\{\mathrm{b}\}$. Take the translational part (8) as an example, and assume $\boldsymbol{M}_p, \boldsymbol{B}_p, \boldsymbol{K}_p$ are all diagonal matrices, we write the model as:

$$\boldsymbol{M}_p \ddot{\boldsymbol{p}}_e + \boldsymbol{B}_p \dot{\boldsymbol{p}}_e + \left(\boldsymbol{I}_3 - \boldsymbol{S}_p\right) \boldsymbol{K}_p \boldsymbol{p}_e = \mathcal{F}_b^f - \boldsymbol{S}_p \mathcal{F}_d^f \tag{22}$$

where $\boldsymbol{S}_p = \mathrm{diag}(\sigma_x^p, \sigma_y^p, \sigma_z^p)$, $\sigma_{x,y,z}^p \in \{0, 1\}$ and $\mathcal{F}_d^f$ is the desired force applied to the environment. $\sigma_{x,y,z}^p$ can be regarded as a selector where 1 means a force control mode and 0 means an impedance mode along the corresponding direction. Note that if the selector matrix $\boldsymbol{S}_p$ is set to be an identity with zero desired force, the model (22) yields a zero-gravity free drag mode. Similarly, for the moment control model, we have

$$\boldsymbol{M}_r \ddot{\boldsymbol{r}}_e + \boldsymbol{B}_r \dot{\boldsymbol{r}}_e + (\boldsymbol{I}_3 - \boldsymbol{S}_r) \boldsymbol{K}_r \boldsymbol{r}_e = \mathcal{F}_b^m - \boldsymbol{S}_r \mathcal{F}_d^m \tag{23}$$

where $\boldsymbol{S}_r = \mathrm{diag}(\sigma_x^r, \sigma_y^r, \sigma_z^r)$, $\sigma_{x,y,z}^r \in \{0, 1\}$ and $\mathcal{F}_d^m$ is the desired moment applied to the environment. The practical controller is employed to implement the force and moment control model due to its improved coupled stability.

G. Gravity and Inertia Force Compensation

The measurement of the wrist force sensor include the gravity and inertial force components of the overload (e.g., US probe), which depend on the pose and motion of the end-effector. Gravity and inertia force compensation have to be performed to obtain the external wrench $\mathcal{F}_b$. Let us denote by $\{\mathrm{c}\}$ the center-of-mass frame of the overload. The fixed transformation matrix between $\{\mathrm{c}\}$ and $\{\mathrm{b}\}$ is given by $\boldsymbol{T}_{cb} = (\boldsymbol{I}_3, -\boldsymbol{p}_{\mathrm{com}})$, where $\boldsymbol{p}_{\mathrm{com}}$ is the position vector of the mass-of-center expressed in $\{\mathrm{b}\}$. Let $\mathcal{I}_c \in \mathbb{R}^{3\times 3}$ and $\mathcal{M}$ represent the inertia matrix and mass, respectively. The dynamics of the overload (single rigid body) expressed in $\{\mathrm{c}\}$ is given by the Newton-Euler equation as

$$\mathcal{F}_c^{\mathrm{total}} = \mathcal{G}_c \dot{\mathcal{V}}_c - \left[\mathrm{ad}_{\mathcal{V}_c}\right]^{\mathrm{T}} \mathcal{G}_c \mathcal{V}_c \tag{24}$$

where $\mathcal{F}_c^{\mathrm{total}}$ is the total wrench acting on the overload; $\mathcal{G}_c = \mathrm{diag}(\mathcal{I}_c, \mathcal{M}\boldsymbol{I}_3) \in \mathbb{R}^{6\times 6}$ is the spatial inertia matrix; $\mathcal{V}_c = (\boldsymbol{\omega}_c; \boldsymbol{v}_c)$ is the twist of the overload expressed in $\{\mathrm{c}\}$; $\left[\mathrm{ad}_{\mathcal{V}_c}\right] \in \mathbb{R}^{6\times 6}$ is defined as

$$\left[\mathrm{ad}_{\mathcal{V}_c}\right] = \begin{pmatrix} [\boldsymbol{\omega}_c] & \boldsymbol{0} \\ [\boldsymbol{v}_c] & [\boldsymbol{\omega}_c] \end{pmatrix} \tag{25}$$

Note that $\mathcal{V}_c = [\mathrm{Ad}_{\boldsymbol{T}_{cb}}]\mathcal{V}_b$, $\dot{\mathcal{V}}_c = [\mathrm{Ad}_{\boldsymbol{T}_{cb}}]\dot{\mathcal{V}}_b$ with $\mathcal{V}_b = \boldsymbol{J}_b \dot{\boldsymbol{q}}$, $\dot{\mathcal{V}}_b = \dot{\boldsymbol{J}}_b \dot{\boldsymbol{q}} + \boldsymbol{J}_b \ddot{\boldsymbol{q}}$, the right side of Eq. (24) are completed determined by the robot's structural parameters (e.g., D-H or POE parameters) and joint state $(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ which can be measured using joint encoders. The left side of Eq. (24) is composed of

$$\mathcal{F}_c^{\mathrm{total}} = -\mathcal{F}_c^{\mathrm{ext}} - \mathcal{F}_c^{\mathrm{sensor}} + \mathcal{F}_c^{\mathrm{gravity}} \tag{26}$$

where $\mathcal{F}_c^{\mathrm{ext}}$ is the external wrench exerted on the environment; $\mathcal{F}_c^{\mathrm{sensor}}$ is the wrench measured by the wrist force sensor; $\mathcal{F}_c^{\mathrm{gravity}} = \left(\boldsymbol{0}; \mathcal{M} g \boldsymbol{R}^{-1} \hat{z}\right)$ is the gravity wrench with $\hat{z} = (0, 0, -1)^{\mathrm{T}}$ being the gravity direction expressed in the robot's

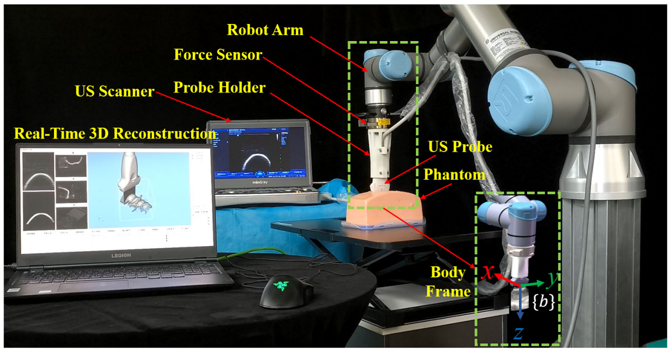

Fig. 6. Experimental scene.

global frame. With replacement of $\mathcal{F}_b^{\text{sensor}} = \left[\mathrm{Ad}_{\boldsymbol{T}_{cb}}\right]^{\mathrm{T}} \mathcal{F}_c^{\text{sensor}}$, the external wrench expressed in {b} is given by

$$\begin{aligned} \mathcal{F}_b &= \left[\mathrm{Ad}_{\boldsymbol{T}_{cb}}\right]^{\mathrm{T}} \mathcal{F}_c^{\text{ext}} \\ &= \left[\mathrm{Ad}_{\boldsymbol{T}_{cb}}\right]^{\mathrm{T}} \left(-\mathcal{F}_c^{\text{total}} - \mathcal{F}_c^{\text{sensor}} + \mathcal{F}_c^{\text{gravity}}\right) \\ &= \left[\mathrm{Ad}_{\boldsymbol{T}_{cb}}\right]^{\mathrm{T}} \left(\mathcal{F}_c^{\text{gravity}} - \mathcal{F}_c^{\text{total}}\right) - \mathcal{F}_b^{\text{sensor}} \end{aligned} \tag{27}$$

III. Experiments and Results

A. Experimental Setup

The experimental setup is shown in Fig 6. The robot manipulator was UR5e (Universal Robot, Denmark). A wrist 6-axis force/torque sensor (Gamma, ATI Industrial Automation, USA) was mounted on the flange of the robot. The nominal resolution of the force sensor is 0.05 N and 0.00125 Nm, respectively. An industrial computer (EPC-P3000, Advantech, China) with an Intel Core i7-8700 CPU and 8 GB memory was used as the external controller of the robot via the real-time data exchange (RTDE) protocol. The operating system was Ubuntu 20.04 with RT-Preempt patch for real-time control. All controllers for experimental evaluation were implemented in C++. The control cycle was 2 ms in all experiments. A portable US scanner (M9T, Mindray Corporation, Shenzhen, China) with a linear 2D US probe (L12-4S) and a sector 2D US probe (C5-1S) was employed as the US imaging system. A laptop computer with a video capture card (TC-UB 530, TCHD, China) was used to acquire temporal US images from the scanner for real-time 3D US reconstruction and rendering [17]. The US probe was held by a 3D printed holder which is further connected to the force sensor. The pose information of the US images was sent to the laptop computer from the robot external controller using the UDP protocol with a frequency of 500 Hz. The UR5e robot has an interface to access joint speeds. The joint accelerations were calculated using numerical differentiation which were further filtered by a moving average filter with a length of 50 points. Unless specified, the admittance model parameters (in SI units) were set as $\boldsymbol{M}_r = \mathrm{diag}(0.1, 0.1, 0.1)$, $\boldsymbol{B}_r = \mathrm{diag}(2, 2, 2)$,$\boldsymbol{K}_r = \mathrm{diag}(10, 10, 2)$, $\boldsymbol{M}_p = \mathrm{diag}(0.25, 0.25, 0.25)$, $\boldsymbol{B}_p = \mathrm{diag}(100, 100, 200)$, $\boldsymbol{K}_p = \mathrm{diag}(300, 300, 300)$. The body frame {b} was set as shown in Fig. 6 with the z-direction aligned with the height of the US image, the y-direction aligned with the width of the US image and the x-direction aligned with the normal of the image plane.

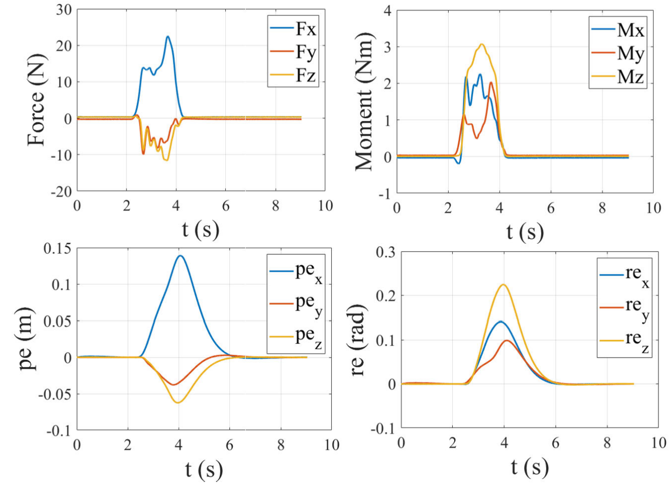

Fig. 7. 6-DoF compliant behavior verification.

B. 6-DoF Compliant Behavior Verification

In order to verify the compliant behavior in task space, external force disturbance is applied to the robotic US probe during the scanning. The robot was controlled with the hybrid admittance controller to move 0.5 m along the x-direction of the body frame {b} at a speed of 0.15 m/s. During the movement, an operator was asked to physically interact with the probe from 2.5 s to 4 s. Fig. 7 shows the change of the interaction wrench (force and moment), and the resulting displacements ($\boldsymbol{r}_e$ and $\boldsymbol{p}_e$) from the desired trajectory. When the external disturbance withdrawn after 4 s, the displacements decayed exponentially to 0, which guaranteed the robot went back to the desired trajectory. Supplementary Video S2 demonstrates the 6-DoF compliant behavior of the robot in the experiment.

C. Compliant Trajectory Following Evaluation

To highlight the advantage of the hybrid admittance controller, compliant trajectory following evaluation was performed. "Compliant trajectory" means that the robot will accordingly deviate from the trajectory when it is subject to external wrench. The "compliance" is guaranteed by the admittance controllers. An online trajectory planner [35] was used to generate a desired trajectory which has a square shape with a side length of 0.5 m. The robot was controlled to follow the desired trajectory with the same orientation in a free space using the practical, ideal and hybrid admittance controllers, respectively. The average linear velocity was 0.15 m/s. The overload of the robot was the US probe and the holder which were 4.8 N in total. The actual trajectories were recorded using an optical tracker with a tracking frequency of 60 Hz. For the hybrid controller, the switch threshold for re-initialization was set to be 1 N. Technically, the hybrid controller and the ideal controller would have the same performance in this experiment since the robot moved in a free space with inertia force and gravity compensation performed. The forementioned trajectory following test was repeated for 5 times. Fig. 8 shows the comparison between the desired and actual trajectories for the three controllers from one trial. The average and maximum

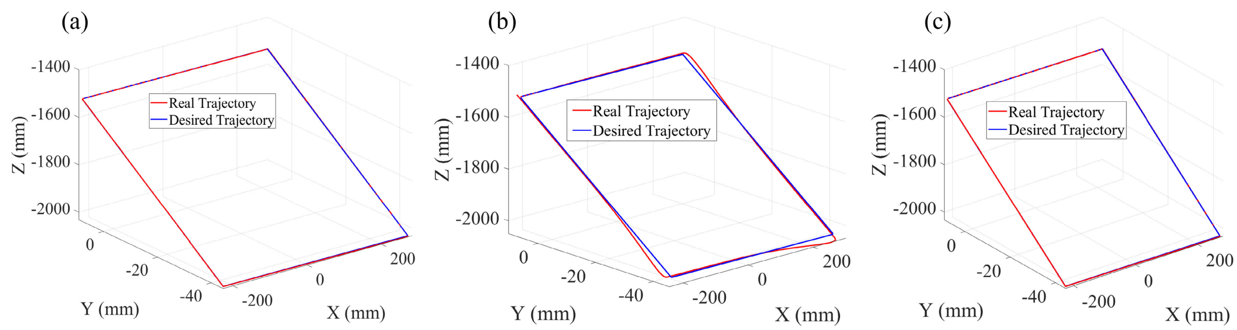

Fig. 8. Compliant trajectory following evaluation. (a) Ideal controller. (b) Practical controller. (c) Hybrid controller.

TABLE II
TRAJECTORY FOLLOWING ACCURACY (UNIT: MM)

Test No.	Ideal		Practical		Hybrid	
	Mean	Max	Mean	Max	Mean	Max
1	0.27	1.28	2.73	18.82	0.26	1.24
2	0.44	1.12	2.35	14.99	0.28	1.22
3	0.44	1.10	2.94	15.11	0.30	1.17
4	0.44	1.12	3.09	15.60	0.30	1.23
5	0.41	1.07	2.94	15.60	0.30	1.21

motion errors represented by the deviation from the desired trajectory are summarized in Table II. The proposed hybrid controller yielded a mean trajectory following error of less than 0.30 mm.

D. Coupled Stability Evaluation

We have shown that the ideal controller has better motion accuracy in free space compared with the practical controller. In Section II-D, we have also analyzed the reason why the practical controller has better coupled stability against the ideal one. In this section, we will experimentally show the comparison when the robot interacts with the human under the three different controllers but with the same model parameters. Actually, in this case the hybrid controller is equivalent to the practical controller since the interaction force exists. An operator was asked to hold the US probe of the robot by a hand and stretch the arm straightly. This would create a motion-constraint environment with interaction force. Then the operator slightly shook the hand to give a small vibration excitation to the probe. The magnitudes of the interaction force and the robot's linear velocity were recorded to check whether oscillation happened which are shown in Fig. 9. The figure clearly indicates the instability of the ideal controller in a motion-constrained environment. Please refer to Supplementary Video S3 for more details.

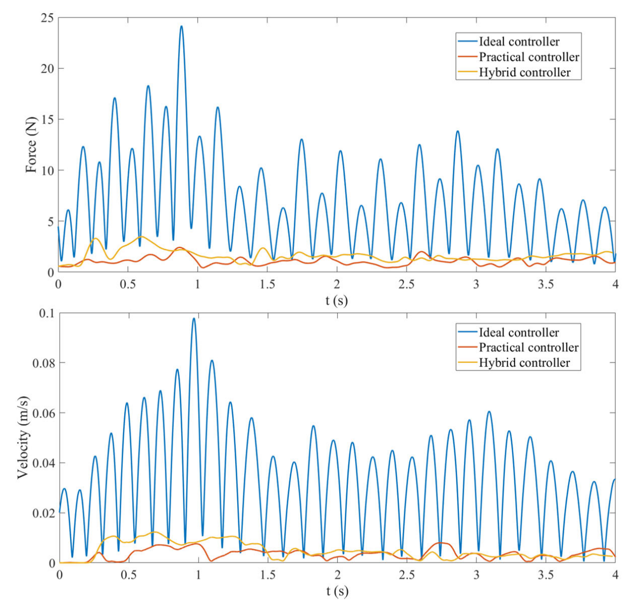

Fig. 9. Coupled stability evaluation. Top figure shows the signal of the magnitude of the interaction force with respect to time and the bottom figure shows the corresponding linear velocity (magnitude) of the robot's end-effector.

TABLE III
CONTACT FORCE CONTROL ACCURACY (UNIT: N)

No.	Error Without Motion		Error With Motion	
	Mean	Max	Mean	Max
1	0.005(0.15%)	0.054(1.81%)	0.014(0.45%)	0.040(1.32%)
2	0.009(0.30%)	0.088(2.92%)	0.008(0.25%)	0.037(1.22%)
3	0.005(0.16%)	0.065(2.17%)	0.008(0.25%)	0.072(2.41%)
4	0.006(0.20%)	0.060(2.00%)	0.015(0.48%)	0.048(1.60%)
5	0.005(0.17%)	0.077(2.58%)	0.006(0.20%)	0.043(1.45%)

E. Force Control Evaluation

1) Contact Force Control Evaluation: We evaluated the contact force control of the robotic US probe using the phantom shown in Fig. 6. The robotic US probe was at its initial position which is above the phantom at a distance of 20 mm. Two types of contact force control evaluation were conducted. First type, the robotic US probe was controlled to push against the phantom surface (z-direction of $\{b\}$) with a contact force of 3 N without lateral motion. Second type, the robotic US probe was controlled to push against the phantom surface (z-direction of $\{b\}$) with a contact force of 3 N and then move along the normal direction of the US image (x-direction of $\{b\}$) driven by a virtual force of 2 N (with lateral motion for scanning). In the other directions of $\{b\}$ the behavior of the probe is compliant. For each type of evaluation, five trials were performed. Fig. 10 shows the contact force

TABLE IV
6-D FORCE CONTROL ACCURACY

No.	F_x(N)		F_y(N)		F_z(N)	
	Mean	Max	Mean	Max	Mean	Max
1	0.012(0.40%)	0.060(1.99%)	0.013(0.43%)	0.063(2.10%)	0.015(0.50%)	0.060(1.99%)
2	0.012(0.40%)	0.060(1.99%)	0.014(0.47%)	0.061(2.03%)	0.012(0.40%)	0.060(1.99%)
3	0.010(0.33%)	0.060(1.99%)	0.012(0.40%)	0.060(1.99%)	0.013(0.43%)	0.063(2.10%)
4	0.012(0.40%)	0.060(1.99%)	0.013(0.43%)	0.060(1.99%)	0.014(0.47%)	0.064(2.13%)
5	0.011(0.37%)	0.060(1.99%)	0.012(0.40%)	0.060(1.99%)	0.011(0.37%)	0.064(2.13%)
No.	M_x(Nm)		M_y(Nm)		M_z(Nm)	
	Mean	Max	Mean	Max	Mean	Max
1	0.0011(0.55%)	0.0042(2.10%)	0.0012(0.60%)	0.0041(2.05%)	0.0012(0.60%)	0.0040(2.00%)
2	0.0013(0.65%)	0.0043(2.15%)	0.0011(0.55%)	0.0040(2.00%)	0.0015(0.75%)	0.0041(2.05%)
3	0.0013(0.65%)	0.0044(2.20%)	0.0013(0.65%)	0.0040(2.00%)	0.0016(0.80%)	0.0041(2.05%)
4	0.0012(0.60%)	0.0043(2.15%)	0.0015(0.75%)	0.0041(2.05%)	0.0012(0.60%)	0.0040(2.00%)
5	0.0012(0.60%)	0.0042(2.10%)	0.0014(0.70%)	0.0040(2.00%)	0.0011(0.55%)	0.0041(2.05%)

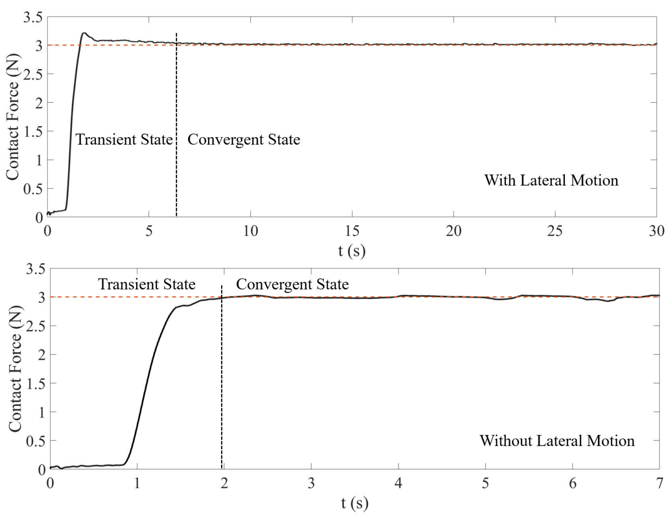

Fig. 10. Contact force control evaluation. Top (bottom) figure shows the transient signal of the regulated contact force without (with) lateral motion.

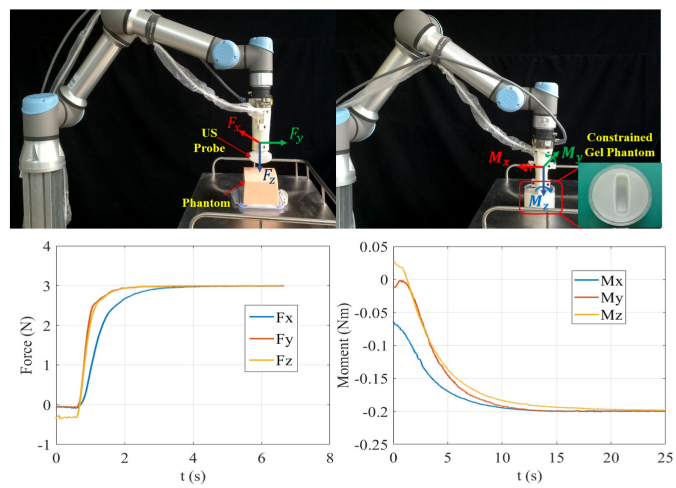

Fig. 11. 6-D force control evaluation. Top figure shows the experimental scene. Bottom figure shows the transient signals of the regulated 6-D force during the evaluation.

signals from one trial of first and second type evaluation, respectively. The actual contact forces were measured by the force sensor and the errors (with relative errors in bracket) compared with the desired force are summarized in Table III. The results show that the relative mean contact force control error was better than 0.5% and the relative max contact force control error was within 3% when the contact force converged.

2) 6-D Force Control Evaluation: We evaluated the control accuracy of the force and moment along the three orthogonal directions of the body frame {b}, respectively. For the three components of force, the robotic US probe was constrained by the phantom along the evaluating direction. Then, the contact force measured by the 6-axis force sensor along the evaluating direction was compared with the desired value to calculate the force control error. For the three components of moment, the robotic US probe was embedded into a gel phantom to create a fully constrained motion-free environment. Then, the moment measured by the 6-axis force sensor along the evaluating direction was compared with the desired value to calculate the moment control error. The desired force and moment were set to be 3 N and 0.2 Nm, respectively. For each direction evaluation, 5 trials were performed. The transient signals of the 6-D force are shown in Fig. 11. The mean and maximum 6-D force control errors (with relative errors in bracket) are summarized in Table IV. The force and moment control mean relative errors are less than 0.5% and 0.8% respectively, and the maximum relative errors are both less than 2.2%. Please refer to Supplementary Video S4 for the experimental process.

F. Automated Robotic US Imaging

To show the feasibility and potential application of our proposed admittance control framework towards automated robotic US imaging, the compliant robotic US imaging system shown in Fig. 6 was evaluated on a human volunteer as shown in Fig. 12. Real-time 3D US reconstruction [17] of the spine, kidney and carotid artery were successfully performed with 6-DoF compliant robotic behavior, as shown in Supplementary Video S5–S7. The physical contact between the robotic probe and the human body was guaranteed by the contact force regulation provided by our control framework. Since the imaging targets are non-rigid and there is no prior 3D geometry available, only qualitative evaluation was performed. A professional

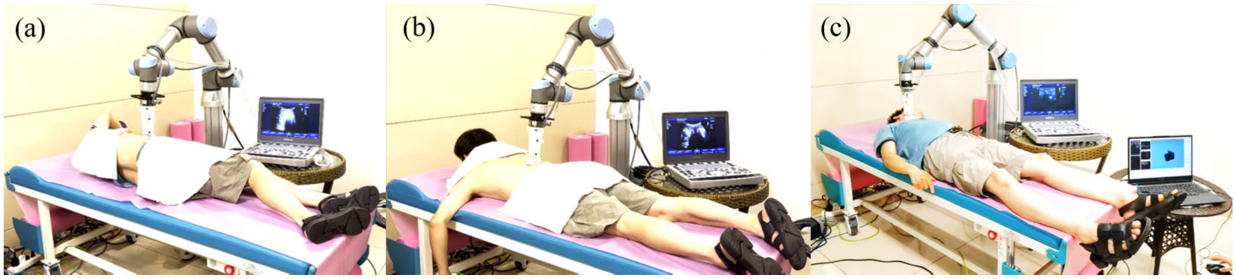

Fig. 12. Automated robotic US imaging on a volunteer. (a) Kidney imaging. (b) Spinal imaging. (c) Carotid artery imaging.

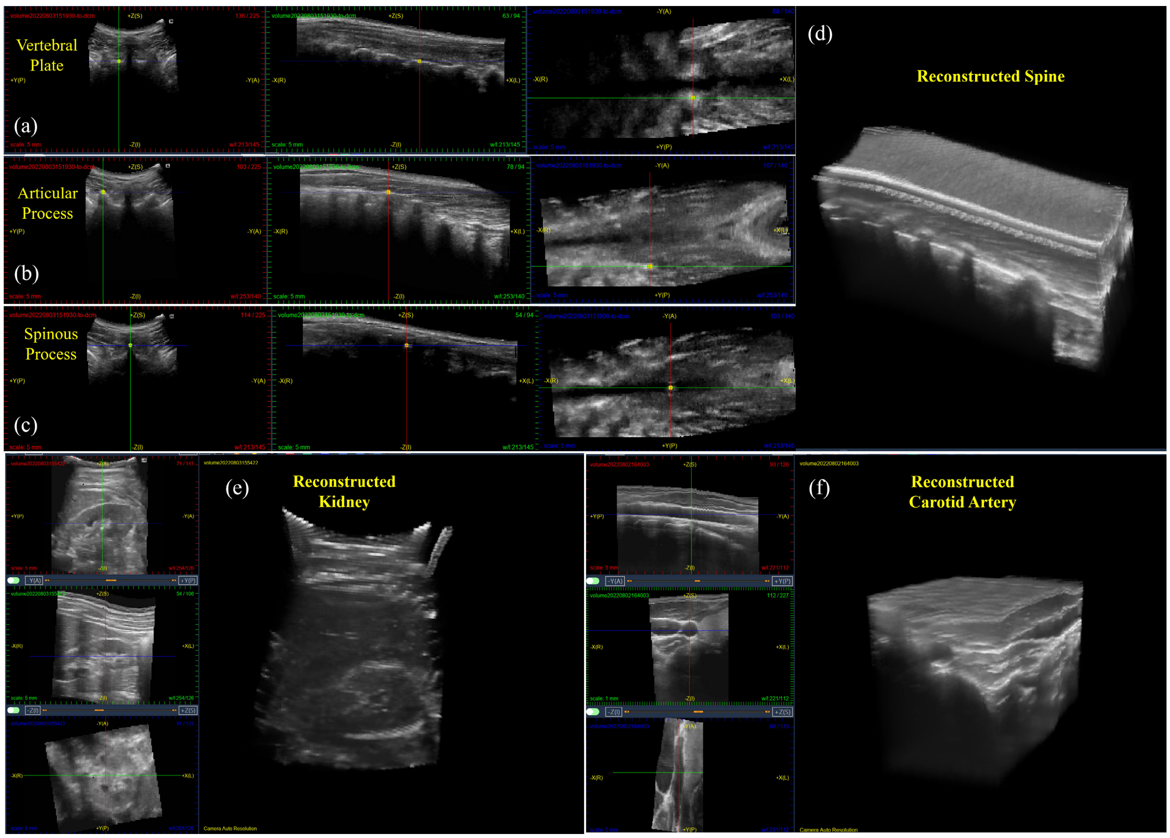

Fig. 13. 3D US reconstruction of a human volunteer. Three kinds of feature points are identified in the 3D reconstruction volume: (a) Vertebral plate; (b) Articular process; (c) Spinous process. (d) 3D reconstruction of the spine. (e) 3D reconstruction of the kidney. (f) 3D reconstruction of the carotid artery.

sonographer was asked to label key structures of the spine in the reconstructed 3D volume. By virtue of the 3D perception, the sonographer could easily identify some key structures such as vertebral plate, articular process and spinous process in the reconstructed 3D US volume, as shown in Fig. 13(a)-(c). Fig. 13(d) shows the reconstructed 3D volume of the spine. Fig. 13(e) and (f) show the sectional view and reconstructed 3D US volume of the kidney and carotid artery. With the measurement utility of the software, the kidney size and radius of the artery could be easily evaluated. Supplementary Video S8 demonstrates the unexpected obstruction of the robotic US probe's motion during the scan and how the compliant robot reacts to this situation for safety.

IV. Conclusion

We have proposed a general admittance control framework for 6-DoF task space compliant motion and 6-D force regulation towards automated robotic US imaging. The framework endows common position-controlled robotic manipulators with the capability of accurate compliant motion in free space and accurate force control when interacts with environment. Three types of controllers, i.e., the ideal, practical and hybrid controllers are proposed. The hybrid controller combines the advantages of the ideal controller which has good accuracy of compliant trajectory following in free space and the practical controller which has improved coupled stability in motion-constrained environment. We have deeply analyzed the reason for different coupled stability behaviors among ideal and practical controllers with the same model parameters. To correctly estimate the external wrench imposed by the robot, a gravity and inertia force compensation approach is proposed. Based on the proposed methods, a 6-DoF compliant automated robotic US imaging system is presented. The 6-D interaction force of the robotic US probe are regulated by the proposed control framework to ensure safe interaction and successful US scanning. Thanks to the accurate kinematics, the 3D poses of the US image sequence are used to construct a real-time 3D US volume to enhance sonographer's spatial perception. Experiments were performed to evaluate the control framework and the automated robotic US imaging system. Experimental results have shown the advantages of the proposed control framework, including satisfied force control accuracy, high accuracy of compliant motion, improved coupled stability, and system effectiveness on a human volunteer.

Appendix

For a rotation matrix $\boldsymbol{R}$, and $\boldsymbol{r} = \log\boldsymbol{R}$, according to [36] we have

$$\frac{\partial \boldsymbol{R}}{\partial r_i} = \frac{r_i[\boldsymbol{r}] + [\boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\boldsymbol{e}_i]}{\|\boldsymbol{r}\|^2}\boldsymbol{R} \tag{A-1}$$

where r_i is the i-th component of $\boldsymbol{r}$ and $\boldsymbol{e}_i$ is the i-th vector of the standard basis of $\mathbb{R}^3$. According to the derivative chain rule, we have

$$\dot{\boldsymbol{R}} = \sum_{i=1}^{3} \frac{\partial \boldsymbol{R}}{\partial r_i}\dot{r}_i \tag{A-2}$$

Substituting Eq. (A-1) into Eq. (A-2) yields

$$\begin{aligned}\dot{\boldsymbol{R}} &= \sum_{i=1}^{3} \frac{r_i[\boldsymbol{r}] + [\boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\boldsymbol{e}_i]}{\|\boldsymbol{r}\|^2}\boldsymbol{R}\dot{r}_i \\ &= \frac{\sum r_i\dot{r}_i[\boldsymbol{r}] + \left[\boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\sum \boldsymbol{e}_i\dot{r}_i\right]}{\|\boldsymbol{r}\|^2}\boldsymbol{R} \\ &= \frac{\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}[\boldsymbol{r}] + \left[\boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\dot{\boldsymbol{r}}\right]}{\|\boldsymbol{r}\|^2}\boldsymbol{R}\end{aligned} \tag{A-3}$$

Note that $[\boldsymbol{\omega}_b] = \boldsymbol{R}^{\mathrm{T}}\dot{\boldsymbol{R}}$ and $\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}$ is a scalar, we have

$$\begin{aligned}[\boldsymbol{\omega}_b] &= \boldsymbol{R}^{\mathrm{T}}\frac{\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}[\boldsymbol{r}] + \left[\boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\dot{\boldsymbol{r}}\right]}{\|\boldsymbol{r}\|^2}\boldsymbol{R} \\ &= \left[\boldsymbol{R}^{\mathrm{T}}\frac{\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}\boldsymbol{r} + \boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\dot{\boldsymbol{r}}}{\|\boldsymbol{r}\|^2}\right]\end{aligned} \tag{A-4}$$

which means

$$\boldsymbol{\omega}_b = \boldsymbol{R}^{\mathrm{T}}\frac{\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}\boldsymbol{r} + \boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})\dot{\boldsymbol{r}}}{\|\boldsymbol{r}\|^2} \tag{A-5}$$

Since $\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}\boldsymbol{r} = \boldsymbol{r}\boldsymbol{r}^{\mathrm{T}}\dot{\boldsymbol{r}}$, we obtain

$$\boldsymbol{\omega}_b = \boldsymbol{R}^{\mathrm{T}}\frac{\boldsymbol{r}\boldsymbol{r}^{\mathrm{T}} + \boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})}{\|\boldsymbol{r}\|^2}\dot{\boldsymbol{r}} \tag{A-6}$$

Comparing with Eq. (10), we have

$$\mathrm{A}(\boldsymbol{r}) = \boldsymbol{R}^{\mathrm{T}}\frac{\boldsymbol{r}\boldsymbol{r}^{\mathrm{T}} + \boldsymbol{r} \times (\boldsymbol{I} - \boldsymbol{R})}{\|\boldsymbol{r}\|^2} \tag{A-7}$$

Replacing $\boldsymbol{R}$ and $\boldsymbol{R}^{\mathrm{T}}$ with

$$\boldsymbol{R} = \boldsymbol{I}_3 + \sin\|\boldsymbol{r}\|\frac{[\boldsymbol{r}]}{\|\boldsymbol{r}\|} + (1 - \cos\|\boldsymbol{r}\|)\frac{[\boldsymbol{r}]^2}{\|\boldsymbol{r}\|^2} \tag{A-8}$$

$$\boldsymbol{R}^{\mathrm{T}} = \boldsymbol{I}_3 - \sin\|\boldsymbol{r}\|\frac{[\boldsymbol{r}]}{\|\boldsymbol{r}\|} + (1 - \cos\|\boldsymbol{r}\|)\frac{[\boldsymbol{r}]^2}{\|\boldsymbol{r}\|^2} \tag{A-9}$$

in Eq. (A-7) and with the following substitutions

$$[\boldsymbol{r}]^2 = \boldsymbol{r}\boldsymbol{r}^{\mathrm{T}} - \|\boldsymbol{r}\|^2\boldsymbol{I} \tag{A-10}$$

$$[\boldsymbol{r}]^3 = -\|\boldsymbol{r}\|^2[\boldsymbol{r}] \tag{A-11}$$

$$[\boldsymbol{r}]^4 = -\|\boldsymbol{r}\|^2[\boldsymbol{r}]^2 \tag{A-12}$$

$$(\sin\|\boldsymbol{r}\|)^2 + (1 - \cos\|\boldsymbol{r}\|)^2 = 2(1 - \cos\|\boldsymbol{r}\|) \tag{A-13}$$

we have the final form of Eq. (11) available.

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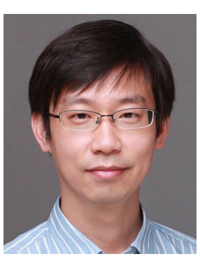

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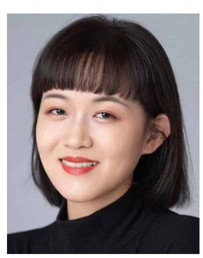

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